# Exit Ticket · reflect and retrieve

## Exit Ticket

Eight-bit two's complement can represent decimal +128.

## Use key vocabulary from memory

1. Explain bit width in your own words.

2. Explain two's complement in your own words.

3. Explain mantissa in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.



RHS COMPUTING / OCR H446 / LESSON 209 DRAFT · TEACHER REVIEW

# Do Now · retrieval

## Exam-style data representation problems

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What does CountAtLeast([5,7,3],5) return?

### Q2

What does O(log n) often describe in binary search?

### Q3

What is the sign bit for a negative 8-bit two's-complement integer?

### Q4

Calculate $0.1000_2 \times 2^3$.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Respect the bit width — Write positive 5 using exactly eight bits.

## Core idea 2

Invert then add one — The result is $11111011_2$, representing −5 in 8-bit two's complement.

## Core idea 3

Interpret floating-point differently — $0.1100_2$ is fractional 0.75; multiplying by $2^3$ gives 6.

**Key visual · complete concept**

**Exam-style data representation problems**

| | | |
|---|---|---|
| **+5** | **00000101** | **Positive** |
| **INVERT** | **11111010** | **All bits** |
| **ADD 1** | **11111011** | **Represents −5** |

## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Normalise positive $0.0011_2 \times 2^5$ using a two's-complement fractional mantissa.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

Convert unsigned binary 00001011 to decimal.

## Question 2

Encode −12 in 8-bit two's complement.

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

bit width — Number of bits in a representation.

two's complement — Fixed-width signed binary format.

mantissa — Signed fractional significant component.

exponent — Power-of-two scaling value.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

Convert −10 to 8-bit two's complement.

## Follow the worked method

1. Predict — Write 00001010 for +10 and invert to 11110101.
2. Explain — Add one: 11110110 represents −10.

**Source model · final illustrated step**

**Convert −10 to 8-bit two's complement.**

| | | |
|---|---|---|
| **+10** | **00001010** | **Start** |
| **INVERT** | **11110101** | **All bits** |
| **ADD 1** | **11110110** | **−10** |

## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict
Start with +7 = 00000111 and invert all bits.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.